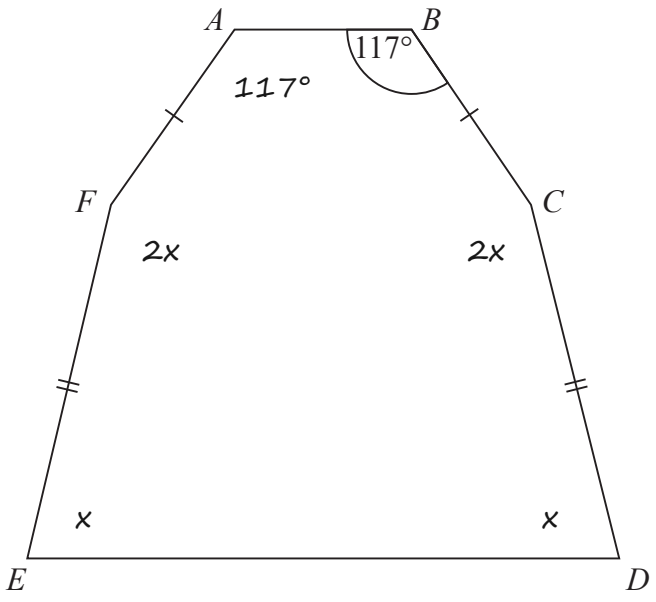
28 The diagram shows a hexagon.
The hexagon has one line of symmetry.



$FA = BC$
$EF = CD$
Angle $ABC = 117°$

Angle $BCD = 2 \times$ angle CDE

Work out the size of angle AFE.
You must show all your working.

Hexagon angle sum = (6 − 2) × 180° = 720°

By symmetry: A = B = 117°, F = C = 2x, E = D = x.

1 mark
Find the hexagon angle sum.

$$234 + 6x = 720$$
$$-234 \qquad -234$$
$$6x = 486$$
$$\div 6 \qquad \div 6$$
$$x = 81$$

1 mark
Use the 2:1 ratio and equal angles.

1 mark
Find x = 81°.

Angle AFE = 2 × 81° = 162°

1 mark
162°, with the working.

162 °

(Total for Question 28 is 4 marks)